



Kenya Certificate of Secondary Education (KCSE)

233/2

CHEMISTRY

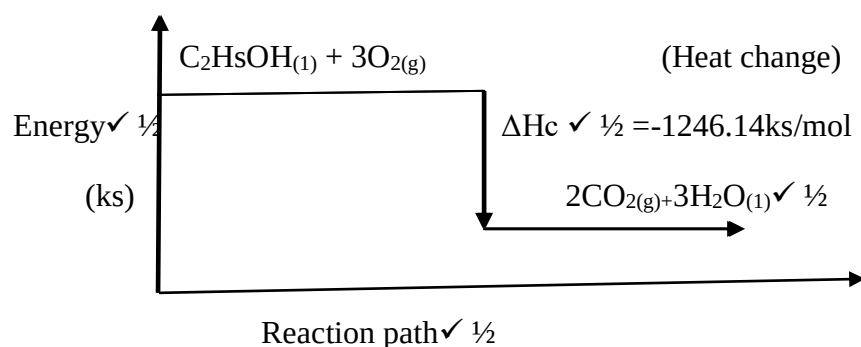
PAPER 2

MARKING SCHEME

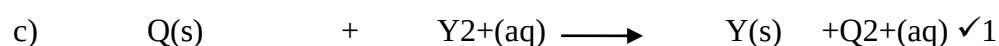
- 1.a) i) Alkali metals ✓ 1
 ii) Minimum energy required to remove an electron from the outermost energy level of an atom ✓ 1 in the gaseous state.
 iii) P has the smallest ✓ 1 atomic radius. Therefore the outermost electron is strongly attracted ✓ 1 to the nucleus hence more energy is required to remove it.
 iv) - It melts because heat is produced ✓ 1 during the reaction or reaction is exothermic.
 - The hissing sound is due to the large production of hydrogen gas ✓ 1 produced in a stream.
 - Moves on the surface of the water due to being propelled ✓ 1 by the escaping hydrogen gas.
 - Floats on surface of the water because it is less denser than water.
 v) $2Q_{(s)} + 2H_2O_{(l)} \longrightarrow 2QOH_{(aq)} + H_{2(g)} \checkmark 1$
 Accept $2Na(s) + 2H_2O(l) \longrightarrow 2NaOH(aq) + H_2(g)$
- b) A strong base is fully ionised ✓ ½ in water to produce many hydroxide ions eg KOH, NaOH or $Na_2O \checkmark \frac{1}{2}$, K_2O while weak base is partially ionised ✓ ½ in water to produce few hydroxide ions eg $NH_3(aq) \checkmark \frac{1}{2}$ / $CaO/Ca(OH)_2/Mg(OH)_2/MgO$
 i) Reaction between a base ✓ 1 and an acid to form salt and water only or reaction between 1 mole of H^+ ions and 1 mole of OH^- to form 1 mole of water.
 ii) Add $200cm^3 \checkmark \frac{1}{2}$ of 2M HNO_3 to $200cm^3$ of 2M Na OH
 - Heat / evaporate the mixture until saturation
 - Allow the mixture to cool for crystals to form slowly
 - Obtain the crystals.
 iii) $2NaNO_3(s) \longrightarrow 2NaNO_2(s) + O_{2(g)}$
2. a) - Cost - Heavy value
 - Availability - Ease of storage
 - Effect on environment - Ease of transportation
- b) i) $DT = 46.5 - 2.5 = 21.5$
 $DH = 450 \times 4.2 \times 21.5$
 $= 40635J \text{ or } 40.635KJ$
 ii) Mass of ethanol = $125.5 - 124 = 1.5g \checkmark \frac{1}{2}$
 RMM of ethanol = $46g \checkmark \frac{1}{2}$
 1.5g of ethanol produces 40635
 46g of ethanol produce $40635 \times \frac{46}{1.5}$
 $= 1246140 \text{ j/mol}$
 $= -1246.14kj/mol$
- c) $C_2H_5OH(l) + 3O_{2(g)} \longrightarrow 2CO_{2(g)} + 3H_2O(l)$



- d) - Heat lost to surrounding air or apparatus
 - Error in reading temperature or mass.
 - Incomplete combustion of ethanol
 - A little of the ethanol evaporates as the burner cools.



3. I. a) $\text{R}^{\frac{1}{2}}$, It has the one $\frac{1}{2}$ with the most negative $E^\ominus$ value.



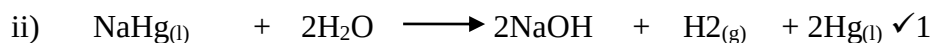
d)
$$\begin{aligned} \text{E.m.f} &= E_{\text{reduced}} - E^\ominus_{\text{oxidised}} \\ &= 0.34 - (-0.13) \\ &= 0.47 \text{ volts.} \end{aligned}$$

e) It has no mobile ions which are required for the reaction with lead to produce electric current.

II) a) i) Water

ii) Titanium

b) Chloride gas



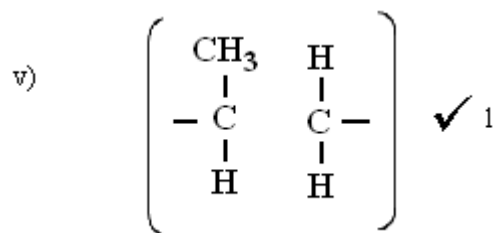
- d) - To minimise cost of operation. It is expensive.
 - To minimise its pollution effects. It is poisonous. (Any 1)

4. a) i) Alkynes

ii) Carboxylic/ Alkanoic acid



- b) i) Vulcanisation ✓ 1
 ii) Hardening the rubber/ to make it stronger or harder.
- c) i) $2\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}(\text{l}) + 2\text{K}(\text{s}) \longrightarrow 2\text{CH}_3\text{CH}_2\text{CH}_2\text{OK} + \text{H}_2(\text{g})$
 ii) I Dehydration
 II Hydrogenation
 iii) A 1,2 – dibromo propane
 B Ethene []
 iv) Nickel/ platinum/palladium



- d) - Source of fuel
 - Manufacture of carbon black used as a component of printers ink
 - Preparation/ manufacture of methanal and chloromethane which are useful industrial chemicals. (Any 2)

5. a) Is the mass of solute ✓ 1 that dissolves in 100g of H₂O or solvent to form a saturation at a given temperature or mass of a solute that saturates 100g of a solvent at a particular/ given temperature.

- b) i) -Scale ✓ 1 (½ each axis)

-Plotting A ✓ 1

B ✓ 1

-Curves A ✓ 1

B ✓ 1

- c) i) $24^\circ\text{C} \pm 1^\circ\text{C}$ ✓ 1
 ii) Solubility at $50^\circ\text{C} = 37\text{g}/100\text{g}$
 $100\text{g} \longrightarrow 37\text{g}$



$$35\text{g} \longrightarrow ? = \left(\frac{35 \times 37}{100} \right) \checkmark \frac{1}{2}$$

$$= 12.95\text{g} \checkmark \frac{1}{2}$$

iii) From graph

Solubility at 50°C of A = 84g/ 100g $\checkmark \frac{1}{2}$

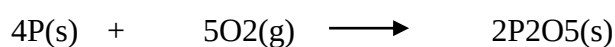
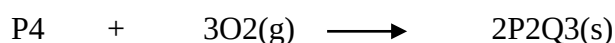
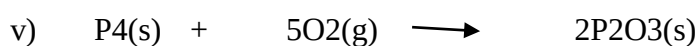
Solubility at 50°C of B = 37g/ 100g

$$84 - 37 = 47 \checkmark \frac{1}{2} \pm 1\text{g} (46\text{g} - 48\text{g})$$

d) Fractional crystallisation. $\checkmark \frac{1}{2}$

6. i) Under water – It readily reacts with oxygen when exposed to air.
- ii) Burning phosphorous caused hot air to expand and increase pressure to push down the water. After cooling the volume of air used for combustion of phosphorous was filled by water rising up the bell Jar.
- iii) The white fumes of the phosphorous (v) oxide or Phosphorous (iii) Oxide dissolved in the water to form a colourless.

iv) $\left(\frac{80 - 64}{80} \right) \times 100\% = 20\%$



vi) Red litmus paper remained red/ unchanged

Blue litmus paper turned to red P₂O₃s dissolved in water to form phosphoric acid

b) i) $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$

ii) Cuts off supply Oxygen

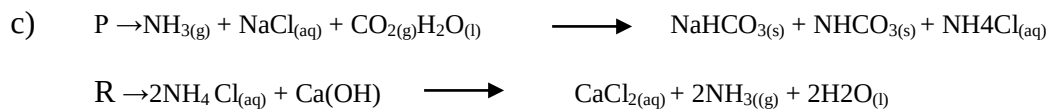
iii) They have a sacrificial metal on them.

7. a) A – Carbon(iv)Oxide

B - Ammonia

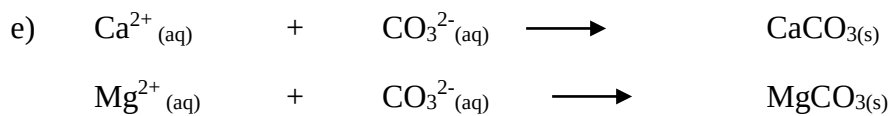
b) C – Ammonium chloride

D – Sodium hydrogen carbonate



d) Calcium chloride – used as a drying agent

Used to lower the melting point of molten NaCl in the extraction of sodium



Ca^{2+} and Mg^{2+} causing the hardness are precipitated as insoluble carbonates.